



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2021 - 2022)

JEE (Advanced)

REVIEW TEST

08-08-2021

JEE(Main + Advanced) : Nurture Course (PHASE : S(TEPS))

ANSWER KEY

PAPER-1

PART-1 : PHYSICS

SECTION-I (i)	Q.	1	2	3	4	5	6		
	A.	A,B,C,D	A,B,C,D	B,C	A,C	B,C,D	A,C		
SECTION-I (ii)	Q.	7	8	9	10				
	A.	A	C	A	A				
SECTION-II	Q.	1	2	3	4	5	6	7	8
	A.	0.10	6.00	500.00	0.00	275.00	1.41	3.00	4.55

PART-2 : CHEMISTRY

SECTION-I (i)	Q.	1	2	3	4	5	6		
	A.	A,B,C	B,C	A,C	A,B,C,D	A,B,C	B,D		
SECTION-I (ii)	Q.	7	8	9	10				
	A.	D	D	B	C				
SECTION-II	Q.	1	2	3	4	5	6	7	8
	A.	2.00	2.00	4.00	2.00	56.00	4.00	7.00	6.00

PART-3 : MATHEMATICS

SECTION-I (i)	Q.	1	2	3	4	5	6		
	A.	A,B,C,D	A,B	A,C,D	B,D	C,D	A,D		
SECTION-I (ii)	Q.	7	8	9	10				
	A.	A	D	D	D				
SECTION-II	Q.	1	2	3	4	5	6	7	8
	A.	5.00	5.00	0.25	8.00	6.00	4.00	4.00	4.00

HINT – SHEET

PART-1 : PHYSICS

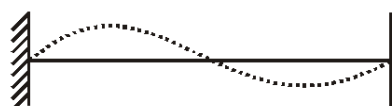
SECTION-I (i)

1. Ans (A,B,C,D)

$$v_p = -u \frac{\partial s}{\partial x}$$

$$\delta_p = -B \frac{\partial s}{\partial x}$$

2. Ans (A,B,C,D)



$$u = A_x \sin \omega t$$

$$A_x = \frac{A}{2} = A \sin kx$$

$$kx = \frac{\pi}{6}, \frac{5\pi}{6}, \frac{7\pi}{6}, \frac{11\pi}{6}$$

$$x = \frac{L}{12}, \frac{5L}{12}, \frac{7L}{12}, \frac{11L}{12}$$

3. Ans (B,C)

$$F(R+r) = \frac{3}{2}mR^2\alpha$$

$$\Rightarrow R\alpha = 2 \frac{F(R+r)}{3mR}$$

$$F + f = ma$$

$$\Rightarrow f = \frac{2F(R+r)}{3mR} - F$$

4. Ans (A,C)

$$f_{av} = \frac{f_1 + f_2}{2}$$

In top f_{av} is more $\Rightarrow$ it is superposition of 300

Hz & unknown.

$$f_b = f_1 - f_2$$

$$f_{btop} = 2\text{Hz} \text{ \& } f_{bbottom} = 4\text{Hz}$$

5. Ans (B,C,D)

$$\ell = 2\pi\sqrt{\frac{m}{k}} \Rightarrow m = \frac{k}{4\pi^2} = \frac{2000}{40}$$

$$mg = kx_0$$

$$\Rightarrow x_0 = \frac{500}{200} = 0.25\text{m}$$

$$k\ell = \text{constant.}$$

$$2000 \times 1 = k' \times \frac{1}{2}$$

$$k' = 4000$$

$$\Rightarrow \text{TP decreases}$$

$$T = 2\pi\sqrt{\frac{50}{4000}} = \frac{\pi}{2\sqrt{5}}$$

6. Ans (A,C)

$$f = \frac{C + \frac{C}{5}}{C - \frac{C}{5}} \times f_0 = 1.5f_0$$

$$\lambda = \frac{C - \frac{C}{5}}{f_0} = \frac{4C}{5f_0}$$

Let as say a packet has energy E_0 same packet

is emitted with a smaller time period.

$$\Rightarrow \frac{E_0}{T} = P_0 \Rightarrow E_0 f_0$$

$$P = \frac{E_0}{T} = E_0 \times 1.5 = 1.5P_0$$

SECTION-I (ii)

7. Ans (A)

$$T_Q = 2\pi\sqrt{\frac{m}{k}}, T_P = 2\pi\sqrt{\frac{m}{4k}} = \frac{1}{2}$$

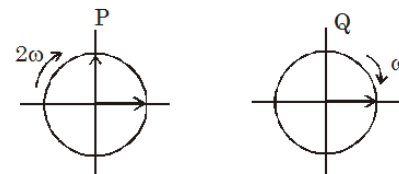
when Q is lowest, P has come back to original position.

$$mgx - \frac{1}{2}kx^2 = 0 \Rightarrow x_{\max} = \frac{2mg}{k}$$

8. Ans (C)

$$v_I - v_M = -(v_0 - v_M)$$

$$\Rightarrow v_I = 2v_M - v_0 = 2v_P - v_Q$$



$$A_P = \frac{mg}{4k} = A, \quad A_Q = \frac{mg}{k} = 4A$$

$$v_P = 2A\omega \sin 2\omega t$$

$$v_Q = 4A\omega \sin \omega t$$

Both are at their upper extreme.

$$\omega = \sqrt{\frac{k}{m}}$$

$$v_I = 4A\omega \sin 2\omega t - 4A\omega \sin \omega t$$

$$\frac{dv_I}{dt} = 0 \Rightarrow 2 \cos 2\omega t - \cos \omega t = 0$$

$$\Rightarrow 2 [2\cos^2 \omega t - 1] - \cos \omega t = 0$$

$$\Rightarrow 4\cos^2 \omega t - \cos \omega t - 2 = 0$$

$$\cos \omega t = \frac{1 \pm \sqrt{1+32}}{8}$$

$$\frac{d^2v_I}{dt^2} = (-4 \sin 2\omega t + \sin \omega t)$$

$$= \sin \omega t [1 - 8 \cos \omega t]$$

$$= -ve$$

SECTION-II

1. Ans (0.10)

$$80 = m\omega^2 x$$

$$= 0.1 \omega^2 \times 0.2$$

$$4000 = \omega^2$$

$$\omega = 20\sqrt{10} = \frac{2\pi}{T}$$

$$T = 0.1 \text{ sec.}$$

2. Ans (6.00)

$$I_1 = \frac{8\pi}{4\pi \times I^2} = 2 \text{ W/m}^2$$

$$I_2 = \frac{18\pi}{4\pi \times \frac{9}{4}} = 2 \text{ W/m}^2$$

$$\Delta\phi = \frac{2\pi}{\lambda} \Delta x = \frac{2\pi}{3} \times 0.5 = \frac{\pi}{3}$$

$$I = I_1 + I_2 + 2\sqrt{I_1 I_2} \cos \Delta\phi$$

$$= 4 + 4 \times \frac{1}{2} = 6 \text{ W/m}^2$$

3. Ans (500.00)

$$v_{\text{water}} = \sqrt{\frac{\beta}{\rho}} = 10^3 \text{ m/s}$$

$$t = \frac{L}{10^3}$$

$$t' = \frac{3L}{1000}$$

$$t' - t = 1$$

$$\frac{2L}{1000} = 1$$

$$L = 500.00 \text{ m}$$

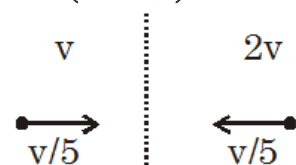
4. Ans (0.00)

$$mg - T - \frac{mg}{2} = ma; B = \frac{\rho \omega v g}{\omega} = \frac{mg}{2}$$

$$\text{and } T - \frac{mg}{2} = ma$$

$$\Rightarrow a = 0$$

5. Ans (275.00)



$$f_1 = \frac{v}{v - \frac{v}{5}} \times 200 = 250 \text{ Hz}$$

$$f_1 = f_{\text{transmitted}} = 250 \text{ Hz}$$

$$f_2 = \frac{2v + \frac{v}{5}}{2v} \times f_1$$

$$= \frac{11}{10} \times 250 = 275 \text{ Hz}$$

6. Ans (1.41)



$$2T \sin \theta = mg$$

$$v \approx \sqrt{\frac{T}{\mu}} = \sqrt{\frac{T\ell}{m}}$$

$$t = \frac{2\ell}{v} = 2\ell \sqrt{\frac{m}{T\ell}}$$

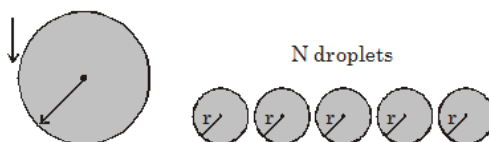
$$= 2\sqrt{\frac{2m\ell}{mg}} \sin \theta$$

$$= 2\sqrt{2 \times \frac{150}{10} \times \frac{3}{\pi} \times \frac{\pi}{180}}$$

$$= 2\sqrt{\frac{150}{300}} = \sqrt{2} \text{ sec}$$

7.

Ans (3.00)



Mass conservation

$$\frac{4}{3}\pi R_c^3 \cdot \rho = N \cdot \frac{4}{3}\pi r^3 \cdot \rho$$

$$R_c^3 = N \cdot r^3$$

$$R_c = N^{1/3} \cdot r$$

Comparison of surface energies :

$$U_B = T \cdot 4\pi R_c^2$$

$$U_S = N \cdot T \cdot 4\pi r^2$$

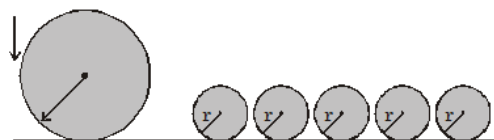
$$\frac{U_B}{U_S} = \frac{1}{N} \left(\frac{R_c}{r} \right)^2$$

$$= \frac{1}{N} \left(N^{1/3} \right)^2 = \frac{1}{N} \cdot N^{2/3}$$

$$\frac{U_B}{U_S} = \frac{1}{N^{1/3}}$$

$$\Rightarrow U_B < U_S$$

Smaller drops combined in to one big droplet in order to decrease the surface potential energy and there is change in gravitational potential energy of bigger drop and droplets, however overall total energy of the system must be conserved.



$$E_S = N \cdot T \cdot 4\pi r^2 + N \left(\rho \cdot \frac{4}{3} \pi r^3 \right) g \cdot r$$

$$E_B = T \cdot 4\pi R_c^2 + \left(\rho \cdot \frac{4}{3} \pi R_c^3 \right) g \cdot R_c$$

8. **Ans (4.55)**

Assume $v_{\text{inside}} = 0$

$$P + 0 + 0 = 1 \text{ atm} + \frac{1}{2} (1000) (30.0)^2$$

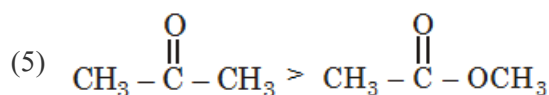
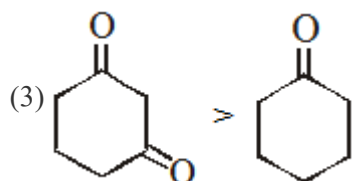
$$+ 1000 (9.80) (0.500)$$

$$P_{\text{gauge}} = P - 1 \text{ atm}$$

$$= 4.50 \times 10^5 + 4.90 \times 10^3 = 455 \text{ kPa}$$

SECTION-II

2. **Ans (2.00)**



8. **Ans (6.00)**



$$x + y = 6$$

PART-3 : MATHEMATICS

SECTION-I (i)

1. **Ans (A,B,C,D)**

$$\text{Let } SA = a, SB = b, \text{ then } \frac{1}{a} + \frac{1}{b} = \frac{1}{p}$$

$$\Rightarrow (a - p)(b - p) = p^2 \Rightarrow a, b = p + 1, p + p^2.$$

$$\text{So, } AB = a + b = p^2 + 2p + 1 = (p + 1)^2$$

$$SP = \sqrt{SA \times SB} = \sqrt{(p + 1)(p + p^2)} = (p + 1) \sqrt{p}$$

$$PA, PB = \sqrt{SP^2 + SA^2}, \sqrt{SP^2 + SB^2}$$

2. **Ans (A,B)**

$$\text{Let } f(x) = (x - \alpha_1)(x - \alpha_2) \dots (x - \alpha_{2018})$$

$$f(0) = f(1)$$

$$\Rightarrow \alpha_1 \times \alpha_2 \dots \alpha_{2018} = (1 - \alpha_1)(1 - \alpha_2) \dots (1 - \alpha_{2018})$$

$$\Rightarrow \alpha_1^2 \times \alpha_2^2 \dots \alpha_{2018}^2 = \alpha_1(1 - \alpha_1) \times \alpha_2(1 - \alpha_2) \dots$$

$$\alpha_{2018}(1 - \alpha_{2018})$$

Apply A.M. $\geq$ G.M.

$$\text{in } \alpha_1 \text{ and } 1 - \alpha_1$$

$$\frac{\alpha_1 + 1 - \alpha_1}{2} \geq (\alpha_1(1 - \alpha_1))^{\frac{1}{2}}$$

$$\Rightarrow (\alpha_1(1 - \alpha_1))^{\frac{1}{2}} \leq \frac{1}{2}$$

$$\Rightarrow (\alpha_1(1 - \alpha_1) \alpha_2(1 - \alpha_2) \dots \alpha_{2018}(1 - \alpha_{2018}))^{\frac{1}{2}} \leq \left(\frac{1}{2}\right)^{2018}$$

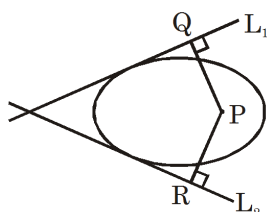
$$\Rightarrow \text{Product of roots} \leq \frac{1}{2^{2018}}$$

3. Ans (A,C,D)

(A) $\because$ Focal chord subtends 90° at directrix
 $\therefore$ Vertex will lie inside circle drawn with ends of focal chord as diameter
 $\therefore$ Focal chord will subtend obtuse angle at vertex.

(B) from fixed point P, perpendiculars PQ & PR are drawn to two fixed lines L_1 & L_2 .

Hence, Q & R are two fixed points lying on auxiliary circle.



So, QR is its fixed chord. Thus, its center will lie on its perpendicular bisector, which is the required locus i.e. a straight line.

(C) $\Delta = a^2|t_1 - t_2||t_2 - t_3||t_3 - t_1| = 8$

$\therefore A.M. \geq G.M.$

$$\Rightarrow \frac{\sum |t_1 - t_2|}{3} \geq \sqrt[3]{8} = 2$$

(But, equality will not hold)

so, $\sum |2t_1 - 2t_2| > 12$

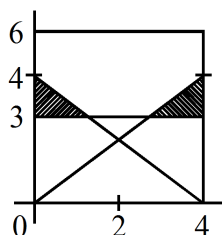
$\Rightarrow \sum |y_1 - y_2| > 12$

(D) The point $P \left(2, \frac{7}{2} \right)$ lies in the region

bounded by an asymptote, tangent at the vertex & the hyperbola

4. Ans (B,D)

Area = $2 \times \left(\frac{1}{2} \times 1 \times 1 \right) = 1$ sq. unit



5. Ans (C,D)

Let $2^x = a$ and $3^x = b$

$$\begin{aligned} \Rightarrow 2^x + 3^x - 4^x + 6^x - 9^x &= a + b - a^2 + ab - b^2 \\ &= -(a^2 - ab + b^2 - a - b + 1) + 1 \\ &= -\frac{1}{2} \left((a-b)^2 + (b-1)^2 + (a-1)^2 \right) + 1 \\ &\leq 1 \end{aligned}$$

6. Ans (A,D)

$$\prod_{i=1}^{99} (x+i) = A_0 + A_1x + \dots + A_{99}x^{99} = \sum_{i=0}^{99} A_i x^i$$

Diff. w.r.t. 'x'

$$\prod_{i=1}^{99} (x+i) \cdot \sum_{i=1}^{99} \frac{1}{x+i} = \sum_{k=1}^{99} k A_k x^{k-1}$$

put $x = 1$

$$100! \sum_{i=1}^{99} \frac{1}{1+i} = \sum_{k=1}^{99} k A_k$$

so, $r_i = 1 + i$

$$S = \sum_{r=1}^{99} \frac{1}{\sqrt{r_k - 1}} = \sum_{i=1}^{99} \frac{1}{\sqrt{i}}$$

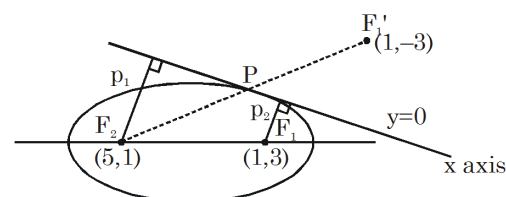
Use

$$2(\sqrt{n+1} - \sqrt{n}) < \frac{1}{\sqrt{n}} < 2(\sqrt{n} - \sqrt{n-1})$$

[S] = 18

SECTION-I (ii)

7. Ans (A)



P, F_1 & F_2 are collinear

$\therefore P(4,0) \Rightarrow 2a = 4\sqrt{2} \Rightarrow a = 2\sqrt{2}$

$p_1 = 1, p_2 = 3$

$\therefore p_1 p_2 = 3 \Rightarrow b^2 = 3$

$e = \sqrt{1 - \frac{3}{8}} = \sqrt{\frac{5}{8}}$

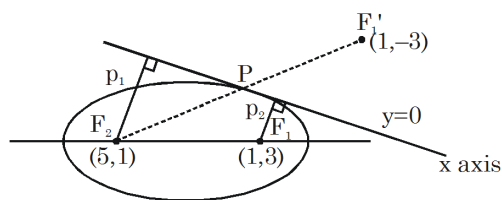
Centre is (3, 2)

Major axis $2y + x = 7$

Minor axis $2x - y = 4$

$\therefore E = 4x^2 + 7y^2 + 4xy - 32x - 40y + 64 = 0$

8. Ans (D)



$P, F_1' \text{ \& } F_2$ are collinear

$$\therefore P(4,0) \Rightarrow 2a = 4\sqrt{2} \Rightarrow a = 2\sqrt{2}$$

$$p_1 = 1, p_2 = 3$$

$$\therefore p_1 p_2 = 3 \Rightarrow b^2 = 3$$

$$e = \sqrt{1 - \frac{3}{8}} = \sqrt{\frac{5}{8}}$$

Centre is (3, 2)

Major axis $2y + x = 7$

Minor axis $2x - y = 4$

$$\therefore E = 4x^2 + 7y^2 + 4xy - 32x - 40y + 64 = 0$$

9. Ans (D)

$$[\sin(\alpha - \beta) + \cos(\alpha + 2\beta)\sin\beta]^2 = 4 \cos\alpha \sin\beta \sin(\alpha + \beta)$$

Use : $\sin(A - B), \sin(A + B)$

$$\cos(A + B)$$

$$(\sin\alpha \cos\beta - \cos\alpha \sin\beta (1 - \cos 2\beta) - 2\sin\alpha \sin^2\beta \cos\beta)^2$$

$$= 4\cos\alpha \sin\beta \sin\alpha \cos\beta + 4\cos^2\alpha \sin^2\beta$$

divide by $\cos^2\alpha \sin^2\beta, \sin\beta \neq 0$

$$\text{we get } (3 - 8\sin^2\beta)^2 = 16$$

$$\sin^2\beta = -1 \text{ or } \frac{7}{8}$$

on check $\sin\beta = 0$ also satisfies system of equation

$$\therefore \sin\beta = \sqrt{\frac{7}{8}}, -\sqrt{\frac{7}{8}}, 0$$

10. Ans (D)

$$[\sin(\alpha - \beta) + \cos(\alpha + 2\beta)\sin\beta]^2 = 4 \cos\alpha \sin\beta \sin(\alpha + \beta)$$

Use : $\sin(A - B), \sin(A + B)$

$$\cos(A + B)$$

$$(\sin\alpha \cos\beta - \cos\alpha \sin\beta (1 - \cos 2\beta) - 2\sin\alpha \sin^2\beta \cos\beta)^2$$

$$= 4\cos\alpha \sin\beta \sin\alpha \cos\beta + 4\cos^2\alpha \sin^2\beta$$

divide by $\cos^2\alpha \sin^2\beta, \sin\beta \neq 0$

$$\text{we get } (3 - 8\sin^2\beta)^2 = 16$$

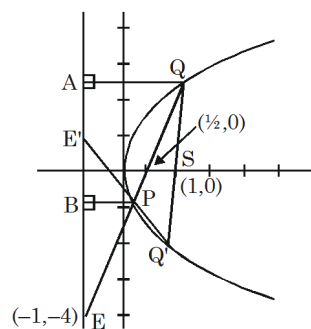
$$\sin^2\beta = -1 \text{ or } \frac{7}{8}$$

on check $\sin\beta = 0$ also satisfies system of equation

$$\therefore \sin\beta = \sqrt{\frac{7}{8}}, -\sqrt{\frac{7}{8}}, 0$$

SECTION-II

1. Ans (5.00)



$$\frac{QS}{SP} = \frac{QA}{PB}$$

$$\frac{QA}{PB} = \frac{EQ}{EP} \quad \{\triangle EAQ \sim \triangle EBP\}$$

$\Rightarrow$ ES is angle bisector of $\angle PSQ'$

similarly E'S is bisector of $\angle PSQ$

Now $y^2 = 4(x - 1)$ and line $8x - 3y - 12 = 0$

$$x - 1 = X, y - 0 = Y$$

$$\therefore Y^2 = 4X \text{ and } 8X - 3Y - 4 = 0$$

$E = (-1, -4)$, equation of E'S is

$$y - 0 = -\frac{1}{2}(x - 1)$$

$$2y + x = 1$$

$$\therefore E'S = (-1, 1)$$

$$\therefore \text{Area of } \Delta = \frac{1}{2} \times 5 \times 2 = 5$$

2. Ans (5.00)

$$\alpha + \beta + \gamma = 0, \alpha\beta + \beta\gamma + \gamma\alpha = -75$$

$$\gamma = -(\alpha + \beta) \Rightarrow \alpha\beta + (\alpha + \beta)\gamma = 75$$

$$\Rightarrow \alpha\beta - (\alpha + \beta)^2 = -75$$

$$\Rightarrow \alpha^2 + \alpha\beta + (\beta^2 - 75) = 0$$

$$\Rightarrow \alpha = \frac{-\beta \pm \sqrt{300 - 3\beta^2}}{2} \dots(1)$$

$$\text{Necessary condition } 300 - 3\beta^2 \geq 0$$

$$\Rightarrow \beta^2 \leq 100 \Rightarrow -10 \leq \beta \leq 10$$

$$\text{but } \beta \in I \text{ also } \alpha \in I$$

$$\therefore \beta = \pm 5, \pm 10 \text{ works in equation (1)}$$

$$\text{if } \beta = 5 \quad (1) \Rightarrow \alpha = 5 \text{ or } -10 \Rightarrow \gamma = -10 \text{ or } 5$$

$$\text{if } \beta = 10 \quad (1) \Rightarrow \alpha = -5 \Rightarrow \gamma = -5$$

$$\therefore |\alpha| + |\beta| + |\gamma| \text{ is constant in each case and is}$$

$$\text{equal to } = 5 + 10 + 5 = 20$$

$$\therefore \left(\frac{|\alpha| + |\beta| + |\gamma|}{4} \right) = 5$$

3. Ans (0.25)

$$(y - 2)^2 = 4 \left(x - \frac{(11 - 3 \sin \theta - 4 \cos \theta)}{4} \right)$$

$$(y - 2)^2 = 4(x - \lambda) \quad \left(\lambda \in \left[\frac{3}{2}, 4 \right] \right)$$

$$\lambda = 4 \text{ due to given constraint}$$

$$\text{Equation of tangent from origin is}$$

$$y - 2 = m(x - 4) + \frac{1}{m}, \text{ this tangent passes}$$

$$\text{through } (0, 0)$$

$$\Rightarrow m = \frac{1 \pm \sqrt{5}}{4}$$

$$\Rightarrow |m_1 m_2| = \frac{1}{4}$$

4. Ans (8.00)

$$(y - \sqrt{2})^2 - 2(x - \sqrt{2})^2 = 4$$

$$\frac{(y - \sqrt{2})^2}{4} - \left(\frac{x - \sqrt{2}}{2} \right)^2 = 1$$

$$b = 2, a = \sqrt{2}, e = \sqrt{1 + \frac{2}{4}} = \sqrt{\frac{3}{2}}$$

$$\therefore \text{Area of } \Delta ABC = \frac{1}{2}(be - b) \left(\frac{2a^2}{b} \right)$$

$$= (e - 1)a^2$$

$$= \left(\sqrt{\frac{3}{2}} - 1 \right) 2 = \sqrt{6} - 2$$

5. Ans (6.00)

$$\text{Let } \tan A = x, \tan B = y, \tan C = z$$

$$\begin{vmatrix} (y+z)^2 & x^2 & x^2 \\ y^2 & (x+z)^2 & y^2 \\ z^2 & z^2 & (x+y)^2 \end{vmatrix} = 2xyz(x+y+z)^2$$

$$= 2 \tan A \tan B \tan C (\tan A + \tan B + \tan C)^3$$

$$= 2(\tan A + \tan B + \tan C)^4 \geq 2 \times 729$$

6. Ans (4.00)

$$\text{Equation of straight line in parametric form is}$$

$$\frac{x+1}{\cos \theta} = \frac{y-2}{\sin \theta} = r$$

$$x = (r \cos \theta) - 1 \text{ \& } y = (r \sin \theta) + 2$$

$$\text{satisfied by the curve}$$

$$(r \cos \theta - 1)(r \sin \theta + 2) = c^2$$

$$r^2 \sin \theta \cos \theta + r(2 \cos \theta - \sin \theta) - 2 - c^2 = 0$$

$$r_1 + r_2 = PA + PB = \frac{\sin \theta - 2 \cos \theta}{\sin \theta \cos \theta} = 2PQ = 2r_3$$

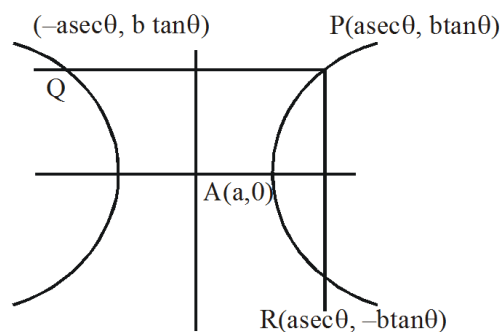
$$2(r_3 \sin \theta)(r_3 \cos \theta) = (r_3 \sin \theta - 2r_3 \cos \theta)$$

$$r_3 \cos \theta = h + 1 \text{ \& } r_3 \sin \theta = k - 2$$

$$2(h + 1)(k - 2) = (k - 2) - 2(h + 1)$$

$$\text{By solving } y = \frac{2x}{(2x + 1)}$$

7. Ans (4.00)



$$(a \sec \theta) (b \tan \theta) = 2ab \tan \theta (\sec \theta - 1)$$

$$\sec \theta = 2 \sec \theta - 2$$

$$\cos \theta = \frac{1}{2}$$

$$\frac{a\pi}{b} = \frac{\pi}{3} \Rightarrow a + b = 4$$

8. Ans (4.00)

Given,

$$(3x - y - 1) + b(4x - y - 2) = 0$$

∴ Fixed point is (1,2)

Let line be

$$(y - 2) = m(x - 1), \quad m \in (-\infty, 0)$$

$$\therefore \text{Area}(m) = \frac{8}{2} + \frac{1}{2} \left(\sqrt{-m} - \frac{2}{\sqrt{-m}} \right)^2, \quad m < 0$$

$$\Rightarrow A_{\text{Least}}(m = -2) = 4$$



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JEE (Advanced)

REVIEW TEST

08-08-2021

JEE(Main + Advanced) : Nurture Course (PHASE : S(TEPS))

ANSWER KEY

PAPER-2

PART-1 : PHYSICS

SECTION-I (i)	Q.	1	2	3	4	5	6		
	A.	A,B,C	B,D	A,D	A,C	A	B,C		
SECTION-I (ii)	Q.	7	8	9	10				
	A.	C	D	C	B				
SECTION-II	Q.	1	2	3	4	5	6	7	8
	A.	3.76 to 3.77	7.07	6.28	2.12	500.00	4.00	1.00	1.00

PART-2 : CHEMISTRY

SECTION-I (i)	Q.	1	2	3	4	5	6		
	A.	B,C,D	A,D	A,B,C,D	B,C,D	A,D	C,D		
SECTION-I (ii)	Q.	7	8	9	10				
	A.	D	D	B	C				
SECTION-II	Q.	1	2	3	4	5	6	7	8
	A.	45.00	51.20	3.00	5.00	8.00	5.00	26.00	17.00

PART-3 : MATHEMATICS

SECTION-I (i)	Q.	1	2	3	4	5	6		
	A.	A,C	A,C	A,B,C,D	B,C	A,C	A,B		
SECTION-I (ii)	Q.	7	8	9	10				
	A.	A	B	C	D				
SECTION-II	Q.	1	2	3	4	5	6	7	8
	A.	1.60	1.20	2.25	6.00	2.50	3.00	5.04	2.80

HINT – SHEET

PART-1 : PHYSICS

SECTION-I (i)

1. Ans (A,B,C)

$$\omega_t = \omega_r = \omega_i ; k_t = 2k_i$$

$$P = \frac{1}{2} \mu v A^2 \omega^2$$

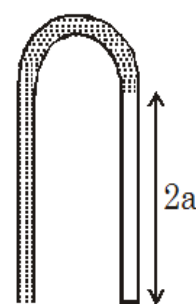
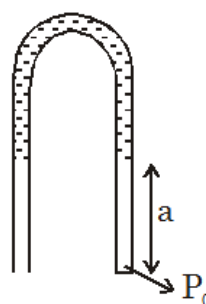
$$A_r = \frac{v_2 - v_1}{v_2 + v_1} \times 6\text{mm}$$

$$v_1 = \frac{\omega}{k} = 100 \text{ m/s}$$

$$v_2 = \frac{v_1}{2} = 50 \text{ m/s}$$

$$A_t = \frac{2v_2}{v_1 + v_2} A_i$$

3. Ans (A,D)



$$P_1 v_1 = P_2 v_2$$

$$P_0 \times aA = (P_0 - 2apg)2at$$

$$2apg = \frac{P_0}{2}$$

$$a = \frac{P_0}{4pg}$$

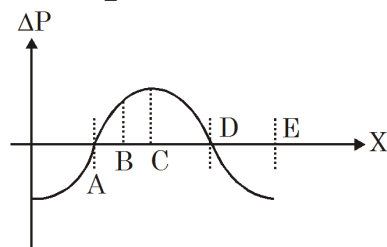
5. Ans (A)

$$\rho = -\frac{Bds}{dx}$$

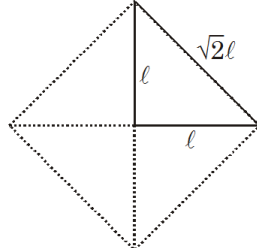
$$v_p = -\frac{vds}{dx}$$

$$v_p = \frac{p}{B}v$$

$$\text{at } \omega t = \frac{\pi}{2}$$

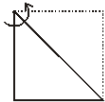


6. Ans (B,C)



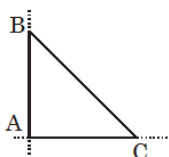
$$4I = \frac{4m(\sqrt{2}\ell)^2}{6}$$

$$\Rightarrow I = \frac{m\ell^2}{3}$$



$$2I_1 = 2m \times \frac{2\ell^2}{3}$$

$$I_1 = \frac{2m\ell^2}{3}$$



$$I_{AB} = I_{AC}$$

$$I_1 = 2I_{AB}$$



$$I_2 = I_{AB} = \frac{I}{2}$$

$$I_3 + I_3 = \frac{2m\ell^2}{12}$$

$$\Rightarrow I_3 = \frac{m\ell^2}{12} = \frac{I}{4}$$

PART-1 : PHYSICS

SECTION-II

1. Ans (3.76 to 3.77)

$$P = Fv = 0.6 \times 0.5 = 0.3 \text{ W}$$

$$I = 60 = 10 \log_{10} \frac{I}{10^{-2}}$$

$$I = 10^{-6} = \frac{P_{\text{rad}}}{4\pi \times 30^2}$$

$$P_{\text{rad}} = 36\pi \times 10^{-4}$$

$$\eta = \frac{36\pi \times 10^{-4}}{0.3} \times 100$$

2. Ans (7.07)

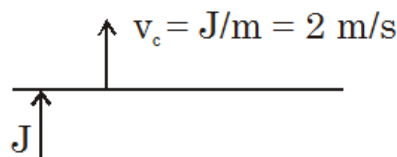
$$\frac{1}{3} m \ell^2 \omega^2 = mg \frac{\ell}{2}$$

$$\omega = \sqrt{\frac{3g}{2\ell}}$$

$$= \sqrt{\frac{3 \times 100}{2 \times 0.3}} = \frac{10}{\sqrt{2}}$$

$$= 5\sqrt{2} = 7.07$$

3. Ans (6.28)



$$J \times \frac{\ell}{2} = I_{\text{cm}} \omega$$

$$4 \times \frac{3}{2} = \frac{2 \times 3}{12}$$

$$\omega = 4 \text{ rad/s}$$

$$t = \frac{4\pi}{\omega} = \pi \text{ sec}$$

$$s = v_c t = 2\pi \text{ m} = 6.28 \text{ m}$$

4. Ans (2.12)

In steady state

$$6\pi\eta r v = mg$$

$$v = \frac{mg}{6\pi\eta r}$$

$$P = mgv$$

$$= \frac{m^2 g^2}{6\pi\eta r}$$

$$= \frac{(20 \times 10^{-3})^2 \times 10^2}{6\pi \times 2 \times 10^{-5} \times \frac{5}{100}}$$

$$= \frac{400 \times 10^{-4}}{6\pi \times 10^{-6}} = \frac{200}{3\pi} \times 10^2$$

PART-2 : CHEMISTRY

SECTION-I (i)

5. Ans (500.00)

$$f = \frac{2}{2 \times 1} \sqrt{\frac{T}{\rho A}} = \sqrt{\frac{y \Delta \ell}{\rho \ell}}$$

$$= \sqrt{\frac{1.98 \times 10^{11} \times 0.01}{8 \times 10^3 \times 0.99}} = 500 \text{ Hz}$$

6. Ans (4.00)

$$\sqrt{2gh} = u_M$$

$$e = 1 = \frac{\sqrt{2gy} - (\sqrt{2gh})}{\sqrt{2gh}}$$

$$y = 4h$$

7. Ans (1.00)

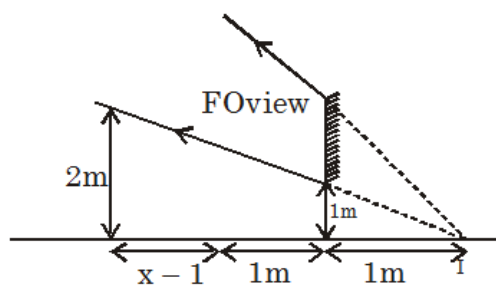
$$\hat{e}_r = \hat{e}_i - 2(\hat{e}_i \cdot \hat{n})\hat{n}$$

$$\vec{e}_r = \vec{e}_i - 2(\vec{e}_i \cdot \hat{n})\hat{n}$$

$$= (3\hat{i} + 4\hat{j} + 4\hat{k}) - 2 \frac{(2\hat{i} + 2\hat{j} + \hat{k})}{9} \times 18$$

$$= \hat{i} + 2\hat{j} + 3\hat{k}$$

8. Ans (1.00)



$$\frac{1}{2} = \frac{1}{x+1}$$

$$\Rightarrow x = 1m$$

2. Ans (A,D)

$$\frac{r_A}{r_B} = \frac{n_A}{n_B} \times \sqrt{\frac{M_B}{M_A}}$$

$$\therefore \frac{(\text{Moles of A coming out initially}) / \Delta t}{(\text{Moles of B coming out initially}) / \Delta t} = \left(\frac{2x}{\frac{M_A}{1x}} \right) \times \sqrt{\frac{M_B}{M_A}}$$

$$\therefore \frac{1}{4} = \frac{2}{1} \left(\frac{M_B}{M_A} \right)^{3/2} \quad \therefore \frac{M_B}{M_A} = \frac{1}{4}$$

$$\therefore \left(\frac{M_B}{M_A} \right) = \left(\frac{1}{8} \right)^{2/3}$$

$$\therefore \text{Ratio of moles} = \frac{n_A}{n_B} = \left(\frac{2x}{\frac{M_A}{1x}} \right) = \frac{2M_B}{M_A}$$

$$= 2 \times \frac{1}{4} = \frac{1}{2}$$

Final mass ratio of gases A and B required = 1 : 32

$$\therefore \frac{(n_A)_f}{(n_B)_f} = \left(\frac{\frac{x}{M_A}}{\frac{32x}{M_B}} \right) = \frac{1}{32} \times \frac{1}{4} = \frac{1}{128}$$

$$\text{Now, } \frac{(n_A)_f}{(n_B)_f} = \frac{(n_A)_i}{(n_B)_i} \left[\sqrt{\frac{M_B}{M_A}} \right]^n$$

$$\therefore \frac{1}{128} = \frac{1}{2} \left(\sqrt{\frac{1}{4}} \right)^n$$

(where n is number of step)

upon solving , $n = 6$

PART-2 : CHEMISTRY

SECTION-I (ii)

7. **Ans (D)**

(A) $P = \text{low}$, $T < T_B$

$\therefore Z < 1$ or $PV \neq nRT$

Gas is more compressible with respect to an ideal gas.

$\therefore T > T_c$, so gas cannot be liquefied at any pressure at given temperature.

$$(V_m)_{\text{real}} < (V_m)_{\text{ideal}}$$

(B) $P = \text{very high}$, $T = T_B$

$\therefore Z > 1$ or $PV \neq nRT$

$\therefore T > T_c$, so gas cannot be liquefied at any pressure at given temperature.

$$(V_m)_{\text{real}} > (V_m)_{\text{ideal}}$$

(C) $P = \text{low}$, $T = T_B$

$\therefore Z > 1$ or $PV \neq nRT$

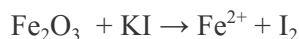
$\therefore T > T_c$, so gas cannot be liquefied at any pressure at given temperature.

PART-2 : CHEMISTRY

SECTION-II

2. **Ans (51.20)**

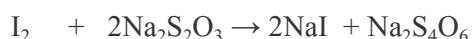
Let $\text{Fe}_2\text{O}_3 = a$ mmole, $\text{Fe}_3\text{O}_4, b = \text{mmole}$



v.f. = 2 v.f. = 2



v.f. = 2 v.f. = 2

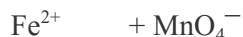


12ml, 0.5 M

3mmole 6mmole

$$a \times 2 + b \times 2 = 3 \times 4 \times 2$$

$$\text{or, } a + b = 12 \quad \dots\dots\dots(1)$$



$$\left[\frac{2a + 3b}{2} \right] \quad \text{v.f.} = 5$$

$$\left[\frac{(2a + 3b)}{2} \right] \times 1 = 12.8 \times 0.25 \times 5$$

$$\text{or, } 2a + 3b = 32 \quad \dots\dots\dots(2)$$

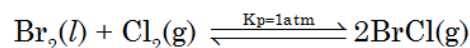
$$\therefore a = 4 \text{ mmole, } b = 8 \text{ mmole}$$

$$\therefore \text{Fe}_2\text{O}_3 = a \text{ mmole} = 4 \text{ mmole}$$

% of $\text{Fe}_2\text{O}_3 =$

$$\frac{4 \times 160 \times 10^{-3}}{1.25} \times 100 = 51.20\%$$

3. **Ans (3.00)**



xmole 1mole 0.1 atm

$$1 = \frac{(0.1)^2}{P_{\text{Cl}_2}} \Rightarrow P_{\text{Cl}_2} \text{ at eq} = 0.01 \text{ atm.}$$

$$\text{Also pressure of Cl}_2 \text{ reacted} = \frac{0.1}{2} = 0.05 \text{ atm}$$

$$\therefore \text{Initial pressure of Cl}_2 = 0.05 + 0.01 = 0.06$$

atm and initial

$$\text{moles of Cl}_2 = 1 \text{ mole} \therefore 0.06 \text{ atm} = 1 \text{ mole}$$

$$\therefore 0.05 \text{ atm} = \frac{1}{0.06} \times 0.05 \text{ mole} = \frac{5}{6} \text{ mole}$$

So $\frac{5}{6}$ moles of Cl_2 is reacted.

$$\therefore \text{moles of Br}_2 \text{ reacted} = \frac{5}{6} \text{ mole.}$$

Also moles of Br_2 corresponding to its vapour

pressure.

$$= \frac{1}{0.06} \times 0.13 = \frac{13}{6} \text{ mole}$$

$$\therefore \text{Minimum moles of Br}_2 = \frac{5}{6} + \frac{13}{6} = \frac{18}{6} = 3 \text{ mole}$$

7. **Ans (26.00)**

Second inert gas configuration =

$$(1s^2 1p^6) (2s^2 1d^{10} 2p^6) \Rightarrow \text{Atomic Number} = 26$$

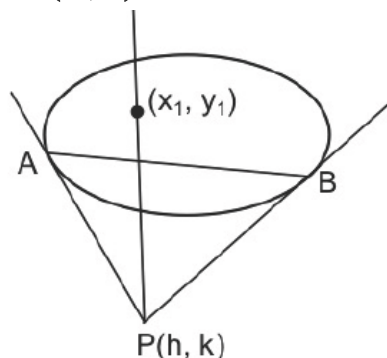
8. **Ans (17.00)**

$$x = 5, y = 12 \Rightarrow x + y = 17$$

PART-3 : MATHEMATICS

SECTION-I (i)

1. Ans (A,C)



Equation of ellipse is $\frac{x^2}{4} + y^2 = p$

Given chord is $x - y = 1$... (i)

chord of contact w.r.t. (h, k) is

$$\frac{xh}{4p} + \frac{yk}{p} = 1 \quad \dots (ii)$$

Comparing (1) and (2) we get

$$\frac{h}{4p} = \frac{k}{-p} = 1$$

$$\Rightarrow h = 4p, k = -p$$

Now reflection of P(h, k) about $x - y = 1$ will be along the line $x + y = \lambda$

Let reflection be (x_1, y_1)

$$\Rightarrow \frac{x_1 + h}{2} - \left(\frac{y_1 + k}{2} \right) = 1,$$

$$\frac{x_1 - y_1 + 5p}{2} = 1$$

$$x_1 - y_1 = 2 - 5p \quad \dots (iii)$$

$$\text{Also, } \frac{y_1 - k}{x_1 - h} = -1$$

$$\Rightarrow x_1 + y_1 = 3p \quad \dots (iv)$$

Solving (3) and (4), we get $x_1 = 1 - p, y_1 = 4p - 1$

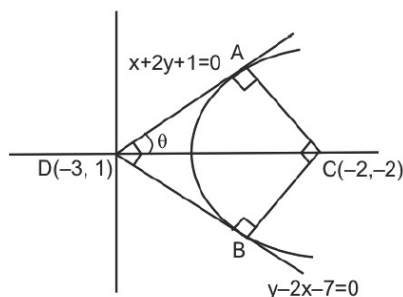
Now (x_1, y_1) has to be inside the ellipse.

$$\Rightarrow (1 - p)^2 + 4(4p - 1)^2 - 4p < 0 \Rightarrow 65p^2 - 38p + 5 < 0$$

$$\Rightarrow 65p^2 - 23p - 25p + 5 < 0 \Rightarrow 13p(5p - 1) - 5(5p - 1) < 0$$

$$\Rightarrow (13p - 5)(5p - 1) < 0 \Rightarrow \frac{1}{5} < p < \frac{5}{13}$$

2. Ans (A,C)



∴ Both tangent are perpendiculars so it will meet at directrix at D(-3, 1).

We know that line joining point of intersection of perpendicular tangent and point of intersection of normal is parallel to axis and we have given that point of intersection of normal (C) lies on axis also D also lies on axis also. So AB will be latus rectum

$$\tan \theta = \left| \frac{3 - \frac{1}{2}}{1 + \frac{3}{2}} \right| = 1 \Rightarrow \theta = 45^\circ.$$

Hence ABCD will be square therefore AB will be latus rectum of the parabola

Hence AB = CD

$$\therefore \text{So equation of axis} \equiv (y - 1) = \frac{3}{-1}(x + 3) \Rightarrow$$

$$y - 1 = -3x - 9 \Rightarrow y + 3x + 8 = 0$$

$$\therefore \text{Length of latus Rectum} = AB = CD (\because \text{ABCD for square}) = \sqrt{(-3 + 2)^2 + (3)^2} = \sqrt{10}$$

3. Ans (A,B,C,D)

Here,

$$(4 + \cos \theta)^2 = 25 + 4 \sin^4 \theta$$

$$\Rightarrow 4 + \cos \theta = 5 \text{ and } \sin \theta = 0$$

$$\Rightarrow \theta = 2\pi \text{ or } \theta = 4\pi \therefore p = 2$$

$$\text{Given } \cos^2 \theta_0 - \frac{1}{4} = \sin^2 \alpha + \frac{\sqrt{3}}{4}$$

$$\Rightarrow \cos^2 \theta_0 - \sin^2 \alpha = \frac{\sqrt{3} + 1}{4}$$

$$\Rightarrow \cos(\theta_0 + \alpha) \cdot \cos(\theta_0 - \alpha) = \frac{\sqrt{3} + 1}{4} \quad \dots (i)$$

$$\text{Now given } (\tan \theta_0 + 1)(\tan \alpha + 1) = 2$$

$$\Rightarrow \tan(\theta_0 + \alpha) = 1 \Rightarrow \theta_0 + \alpha = \frac{\pi}{4}$$

Put in (i)

$$\Rightarrow \cos(\theta_0 - \alpha) = \frac{\sqrt{3} + 1}{2\sqrt{2}}$$

$$\Rightarrow \theta_0 - \alpha = \frac{\pi}{12}$$

$$\Rightarrow \theta_0 = \frac{\pi}{6}, \alpha = \frac{\pi}{12}$$

4. Ans (B,C)

Product of perpendicular from two foci on any tangent = $b^2 = \frac{3}{\sqrt{2}} \cdot \frac{1}{\sqrt{2}} = \frac{3}{2} \Rightarrow b = \sqrt{\frac{3}{2}}$

$$\text{Now } ae = 1 \Rightarrow a^2 = b^2 + a^2e^2 \Rightarrow a = \sqrt{\frac{5}{2}}$$

We know that tangent and normal bisect the angle between focal distances of a point.

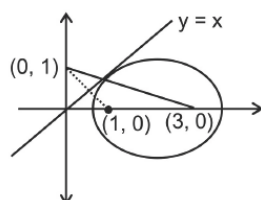


Image of (1, 0) in $y = x$ is (0, 1), line joining (0, 1) & (3, 0) is $x + 3y = 3$. Point of contact of $y = x$ & ellipse

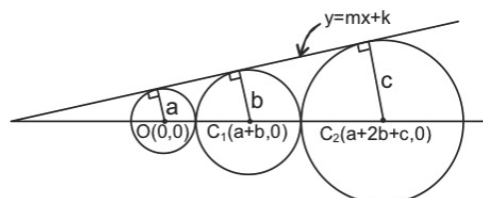
is the point of intersection of $y = x$ and $x + 3y = 3$, i.e. $\left(\frac{3}{4}, \frac{3}{4}\right)$

5. Ans (A,C)

Centre of $C_1 = (a+b, 0)$

Centre of $C_2 = (a+2b+c, 0)$

Let, $y = mx+k$ be common tangent to all three circle



As the line is tangent to the 3 circles,

$$\frac{k}{\sqrt{1+m^2}} = \pm a; \quad \frac{m(a+b)+k}{\sqrt{1+m^2}} = \pm b \quad \& \quad \frac{m(a+2b+c)+k}{\sqrt{1+m^2}} = \pm c$$

As the centres of the 3 circles lie on same side of line, take same sign (+ve) in all three relations, we get,

$$\frac{k}{\sqrt{1+m^2}} = a = b - m(a+b) = c - m(a+2b+c)$$

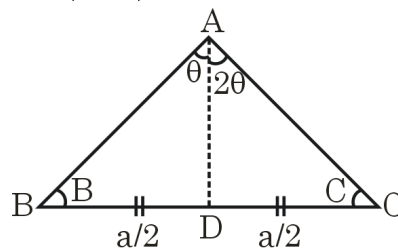
$$\Rightarrow \frac{b-a}{a+b} = \frac{c-a}{a+2b+c} \quad (\text{Eliminating } m)$$

$$\Rightarrow (a+b)(c-a) = (b-a)(a+2b+c)$$

$$\Rightarrow ac - a^2 + bc - ab = ba + 2b^2 + bc - a^2 - 2ab - ac$$

$$\Rightarrow 2b^2 = 2ac \Rightarrow b^2 = ac$$

6. Ans (A,B)



Here, in $\triangle ABD$

$$\frac{AD}{\sin B} = \frac{a}{2 \sin \theta} \quad \dots(i)$$

In $\triangle ADC$,

$$\frac{AD}{\sin C} = \frac{a}{2 \sin 2\theta} \quad \dots(ii)$$

$\therefore$ from (i) & (ii),

$$\frac{a \sin B}{2 \sin \theta} = \frac{a \sin C}{2 \sin 2\theta}$$

$$\therefore 2 \cos \theta = \frac{\sin B}{\sin C} \quad \dots(iii)$$

But $2 \cos \theta = \cot B$

$$\therefore \frac{\cos B}{\sin B} = \frac{\sin C}{\sin B}$$

$$\Rightarrow \cos B = \sin C$$

$$\Rightarrow B = \frac{\pi}{2} - C$$

$$\therefore \boxed{B + C = \frac{\pi}{2}} \quad \dots(iv)$$

$$A + B + C = \pi$$

$$\therefore A = \frac{\pi}{2}$$

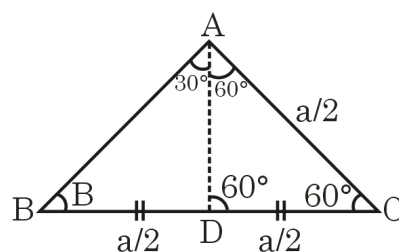
$$A + \theta + 2\theta = \frac{\pi}{2}$$

$$\Rightarrow \theta = \frac{\pi}{6}$$

Now from (i),

$$\frac{AD}{\sin B} = \frac{a}{2 \sin \frac{\pi}{6}}$$

$$\therefore AD = a \sin B = a \cos C$$



Here, In $\triangle ADC$,

$$\text{If } AC = \frac{a}{2}$$

$$\Rightarrow \angle ADC = 60^\circ$$

$\Rightarrow \triangle ADC$ is an equilateral triangle.

Also if $C = 60^\circ \Rightarrow B = 30^\circ$

$\therefore \triangle ABD$ is an isosceles triangle.

PART-3 : MATHEMATICS

SECTION-I (ii)

7. Ans (A)

(P) α, β are roots of $x^2 - kx + 36 = 0$

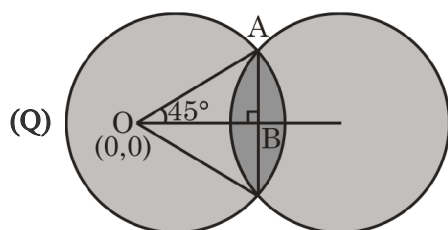
$$k = \alpha + \beta$$

$$\alpha\beta = 1 \times 36 = 2 \times 18 = 3 \times 12$$

$$= 4 \times 9 = 6 \times 6 = (-) \times (-36)$$

$$= \dots (-6) \times (-6)$$

Number of values of $k = 10$



$$S_1 - S_2 = 0$$

$$x - \frac{a}{2} = 0$$

$$OB = \frac{|a|}{2}$$

$$OA = 2\sqrt{2}$$

$$\cos 45^\circ = \frac{|a|}{2 \times 2\sqrt{2}}$$

$$\Rightarrow |a| = 4 = a \pm 4$$

(R) Put $a = 1, b = -1, c = 1$

we get $k = 6$

(S) $\tan 9 - \tan 27 - \tan 63 + \tan 81$

$$(\tan 9 + \tan 81) - (\tan 27 + \tan 63)$$

$$\frac{1}{\cos 9 \cos 81} - \frac{1}{\cos 27 \cos 63}$$

$$\frac{2}{\sin 18} - \frac{2}{\sin 54}$$

$$\frac{8}{(\sqrt{5} - 1)} - \frac{8}{(\sqrt{5} + 1)} = \frac{8 \times 2}{4} = 4$$

8. Ans (B)

$$(P) (1 - \tan^3 \theta) (1 + \tan^2 \theta) + 2^{\tan^2 \theta} = 0$$

$$\text{Put } \tan^2 \theta = t, t \in [0, \infty)$$

$$2^t = t^2 - 1$$

2 value of t and 4 values of θ we get

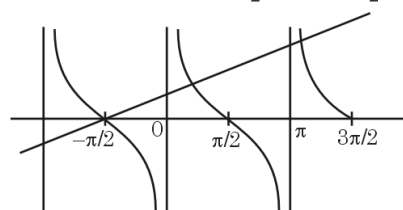
$$(Q) 2x - 3 = 5i$$

$$4x^2 - 12x + 9 = -25$$

$$\Rightarrow 2x^2 - 6x + 17 = 0$$

$$x(2x^2 - 6x + 17) + 4(2x^2 - 6x + 17) + 4$$

$$(R) \cot x = \frac{\pi}{2} + x, x \in \left[-\pi, \frac{3\pi}{2}\right]$$



No. of solution = 3

$$(S) 4\cos x + 3\sin x = 2\lambda + 1$$

$$2\lambda + 1 \in [-5, 5]$$

$$2\lambda \in [-6, 4] \Rightarrow \lambda \in [-3, 2]$$

9. Ans (C)

$$(P) (a,a) \quad (h,k) \quad (b,2b)$$

$$a + b = 2h, a + 2b = 2k$$

$$b = 2(k - h)$$

$$a = 4h - 2k$$

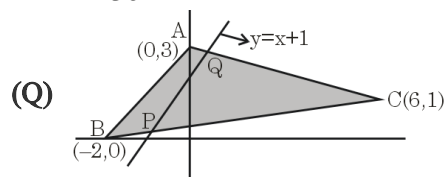
$$(a - b)^2 + (a - 2b)^2 = 16$$

$$(6h - 4k)^2 + (8h - 6k)^2 = 16$$

$$100x^2 + 52y^2 - 144xy = 4 \times 4$$

$$25x^2 + 13y^2 - 36xy = 4$$

$$\lambda = 36$$



$$AC : (y - 3) = -\frac{1}{3}(x - 0) \Rightarrow x + 3y = 9$$

$$Q : x + 3y = 9 \text{ \& } y = x + 1$$

$$x + 3(x + 1) = 9 \Rightarrow x = \frac{3}{2}$$

$$BC : (y - 0) = \frac{1}{8}(x + 2) \Rightarrow 8y = x + 2$$

$$P : y = x + 1 \text{ \& } 8y = x + 2$$

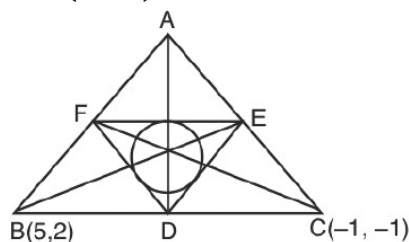
$$8x + 8 = x + 2 \Rightarrow 7x + 6 = 0 \Rightarrow x = -\frac{6}{7}$$

$$\lambda \in \left(-\frac{6}{7}, \frac{3}{2}\right), \lambda = 0, 1 \text{ (two values)}$$

PART-3 : MATHEMATICS

SECTION-II

1. Ans (1.60)



Clearly incentre of $\triangle DEF$ is the orthocentre of $\triangle ABC$

Now (α, α) lies on hyperbola

$\alpha = 3$ $H = (3, 3)$ = incentre of $\triangle DEF$

Now orthocentre of $\triangle BCH$ is A so $A(2, 5)$

$\triangle ABC$ is isosceles so F is the mid point of AB

$$F = \left(\frac{7}{2}, \frac{7}{2}\right) \text{ and } E = \left(\frac{7}{5}, \frac{19}{5}\right)$$

equation of EF = $x + 7y = 28$

$$r = \text{length of } \perp \text{ from } H(3, 3) \text{ on } FE = r = \frac{2\sqrt{2}}{5}$$

2. Ans (1.20)

Any point on line

$(\alpha + r \cos \theta, 2 + r \sin \theta)$ meet the ellipse

$$\frac{x^2}{9} + \frac{y^2}{4} = 1 \text{ at A and D}$$

$$\frac{(\alpha + r \cos \theta)^2}{9} + \frac{(2 + r \sin \theta)^2}{4} = 1$$

$$(4\cos^2\theta + 9\sin^2\theta)r^2 + 4(2\alpha \cos\theta + 9 \sin\theta) r + 4\alpha^2 = 0$$

$$PA \cdot PD = \frac{4\alpha^2}{4\cos^2\theta + 9\sin^2\theta}$$

Again solving with $xy = 0$

$$PB \cdot PC = \frac{2\alpha}{\sin \theta \cos \theta}$$

$$PA \cdot PD = PB \cdot PC$$

$$\frac{4\alpha^2}{4\cos^2\theta + 9\sin^2\theta} = \frac{2\alpha}{\sin \theta \cos \theta}$$

$$\Rightarrow 2\alpha = 4\cot\theta + 9\tan\theta$$

$$\Rightarrow 2\alpha \sin\theta \cos\theta = 4\cos^2\theta + 9\sin^2\theta$$

$$\Rightarrow 2\alpha \sin 2\theta + 5\cos 2\theta = 13$$

$$\Rightarrow -1 \leq \frac{13}{\sqrt{4\alpha^2 + 25}} \leq 1$$

$$\Rightarrow |\alpha| \geq 6$$

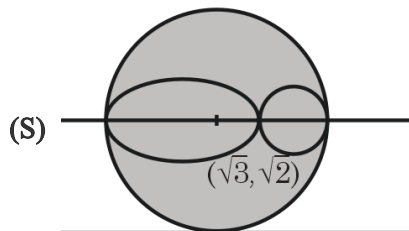
(R) $(x + y + C_1)(x + y + C_2) = 0$

$$25\sqrt{2} = \left| \frac{C_1 - C_2}{\sqrt{2}} \right| \Rightarrow |C_1 - C_2| = 50$$

$$C_1 + C_2 = -8a, \quad C_1 C_2 = -9a^2$$

$$(C_1 + C_2)^2 - 4C_1 C_2 = 2500$$

$$64a^2 + 36a^2 = 2500 \Rightarrow |a| = 5$$



(S)

$$2r_1 + 2r_2 = 4$$

$$r_1 + r_2 = 2$$

10. Ans (D)

(P) $x = \sqrt{31 + x} \Rightarrow x^2 - x - 31 = 0$

$$x = \frac{1 + \sqrt{125}}{2}$$

$$y = \sqrt{1 + y} \Rightarrow y^2 - y - 1 = 0$$

$$\Rightarrow y = \frac{1 + \sqrt{5}}{2}$$

$$\frac{x}{y} = \frac{1 + \sqrt{125}}{1 + \sqrt{5}} = \frac{1 + (\sqrt{5})^3}{1 + \sqrt{5}} = 6 - \sqrt{5}$$

(Q) $m = \frac{(n-4)(n+3) - 23}{n-4}$

$$m = n + 3 - \frac{23}{n-4}$$

$$n-4 = \pm 1, \pm 23$$

$$\Rightarrow n = 5, 3, 27, -19$$

only value of $n = 27$ for which $m \in \mathbb{N}$

$$m = 30 - 1 = 29$$

(R) $x = \frac{-\log 5}{\log 3} \cdot \frac{3 \log 7}{3 \log 5} \cdot \frac{6 \log 3}{2 \log 7} = -3$

$$y = 25^{\frac{3 \log 11}{2 \log 17} \cdot \frac{\log 17}{2 \log 28} \cdot \frac{2 \log 28}{3 \log 11}} = 25^{\frac{1}{2}} = 5$$

$$x^2 + y^2 = 34$$

(S) $|x| = t$

$$|3t - 2| \geq |2t - 2|$$

$$9t^2 - 12t + 4 \geq 4t^2 - 8t + 4$$

$$5t^2 - 4t \geq 0$$

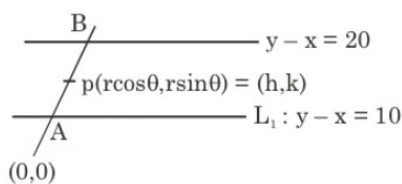
$$\Rightarrow t \in (-\infty, 0] \cup \left[\frac{4}{5}, \infty\right)$$

$$|x| \leq 0 \text{ \& \; } |x| \geq \frac{4}{5}$$

$$x = 0 \text{ \& \; } x \in \left(-\infty, -\frac{4}{5}\right] \cup \left[\frac{4}{5}, \infty\right)$$

$$x \in \left(-\infty, -\frac{4}{5}\right] \cup \left[\frac{4}{5}, \infty\right) \cup \{0\} - \{-1, 1\}$$

3. Ans (2.25)



Let $OA = r_1$, $OB = r_2$

$$r_1 = \frac{10}{\sin \theta - \cos \theta}, A \equiv (r_1 \cos \theta, r_1 \sin \theta)$$

$$r_2 = \frac{20}{\sin \theta - \cos \theta}, B \equiv (r_2 \cos \theta, r_2 \sin \theta)$$

$$\frac{1}{r_2} = \frac{(\sin \theta - \cos \theta)^2}{100} + \frac{(\sin \theta - \cos \theta)^2}{400}$$

$$\frac{(k-h)^2}{100} + \frac{(k-h)^2}{400} = 1 \Rightarrow (k-h)^2 = 80$$

$$\Rightarrow (y-x)^2 = 80$$

4. Ans (6.00)

$$x + y = 2(t^2 + 1)$$

$$y - x = 2t$$

$$\left(\frac{x+y}{2} - 1 \right) = \left(\frac{y-x}{2} \right)^2$$

$$\left(\frac{x+y-2}{2} \right) = \frac{(x-y)^2}{4}$$

$$(x-y)^2 = 2(x+y-4)$$

$$\text{axis : } x - y = 0$$

$$\text{Tangent at vertex : } x + y = 2$$

$$\text{Vertex} \equiv (1, 1)$$

5. Ans (2.50)

$$\theta = \frac{n\pi}{7}$$

$$4\theta + 3\theta = n\pi$$

$$\tan 4\theta = -\tan 3\theta$$

$$\frac{4 \tan \theta - 4 \tan^3 \theta}{1 - 6 \tan^2 \theta + \tan^4 \theta} = -\frac{3 \tan \theta - \tan^3 \theta}{1 - 3 \tan^2 \theta}$$

$$\tan^6 \theta - 21 \tan^4 \theta + 35 \tan^2 \theta - 7 = 0$$

$$\text{put } \tan^2 \theta = z$$

$$z^3 - 21z^2 + 35z - 7 = 0$$

$$\text{put } z^2 = t$$

$$t^3 - 21t^2 + 35t - 7 = 0$$

$$\text{Root are } \tan^2 \frac{\pi}{7}, \tan^2 \frac{2\pi}{7}, \tan^2 \frac{3\pi}{7}$$

$$\tan^2 \frac{\pi}{7} + \tan^2 \frac{2\pi}{7} + \tan^2 \frac{3\pi}{7} = 21$$

$$\text{put } z = \frac{1}{y}$$

$$7y^6 - 35y^4 + 21y^2 - 1 = 0$$

$$\cot^2 \frac{\pi}{7} + \cot^2 \frac{2\pi}{7} + \cot^2 \frac{3\pi}{7} = 5$$

$$\left(\tan^2 \frac{\pi}{7} + \tan^2 \frac{2\pi}{7} + \tan^2 \frac{3\pi}{7} \right) \left(\cot^2 \frac{\pi}{7} + \cot^2 \frac{2\pi}{7} + \cot^2 \frac{3\pi}{7} \right)$$

$$= 21 \times 5 = 105$$

6. Ans (3.00)

$$\alpha + \beta = p \text{ \& } \alpha\beta = q$$

$$\gamma + \delta = q \text{ \& } \gamma\delta = p$$

$$\Rightarrow \alpha + \beta - \alpha\beta + \gamma + \delta - \gamma\delta = 0$$

$$\therefore (\alpha - 1)(\beta - 1) + (\gamma - 1)(\delta - 1) - 2 = 0$$

$$(\alpha - 1)(\beta - 1) + (\gamma - 1)(\delta - 1) = 2$$

$$\text{Case-1 let } (\alpha - 1)(\beta - 1) = 0 \text{ and } (\gamma - 1)(\delta - 1) = 2$$

$$(\gamma, \delta) = (3, 2) \text{ \& } (2, 3) \quad (p, q) = (6, 5)$$

$$\text{Case-2 } (\alpha - 1)(\beta - 1) = 1 \text{ \& } (\gamma - 1)(\delta - 1) = 1$$

$$(\alpha, \beta) = (2, 2) \Rightarrow (p, q) = (4, 4)$$

$$\text{Case-3 } (\alpha - 1)(\beta - 1) = 2 \text{ \& } (\gamma - 1)(\delta - 1) = 0$$

$$(\alpha, \beta) = (2, 3) \Rightarrow (p, q) = (5, 6)$$

$$\text{Hence number of ordered pair} = 3$$

7. Ans (5.04)

$$f(1) = 1, f(2) = -5, f(3) = -9, f(4) = -11$$

$$f(5) = -11, f(6) = -9, f(7) = -5$$

$$f(x) = x^2 - 9x + 9 \text{ true for } x = 1, 2, 3, \dots, 7$$

$$f(x) - (x^2 - 9x + 9) = (x - 1)(x - 2) \dots (x - 7)$$

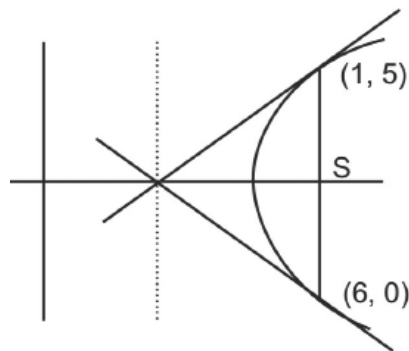
put $x = 8$

$$f(8) - (64 - 72 + 9) = 7.6 \dots 1 = 5040$$

$$f(8) = 5041$$

$$\frac{|f(8)|}{1000} = 5.04$$

8. Ans (2.80)



Point of intersection of tangents at the end of focal chord will lie on the Directrix

Point of intersection of $3x + y = 8$ and $x + 2y = 6$ is $(2, 2)$ which lies on directrix. Foot of $(2, 2)$ on line segment joining $(1, 5)$ and $(6, 0)$ is $(3, 3) =$ focus.

Distance between $(1, 5)$ and $(3, 3)$ is $2\sqrt{2}$

Distance between $(6, 0)$ and $(3, 3)$ is $3\sqrt{2}$

Semi latus rectum is the harmonic mean of segments of focal chord.

$$\Rightarrow \text{Semi latus rectum} = \frac{2 \times 2\sqrt{2} \times 3\sqrt{2}}{2\sqrt{2} + 3\sqrt{2}} = \frac{24}{5\sqrt{2}}$$

$$\Rightarrow \text{latus rectum} = \frac{24\sqrt{2}}{5}$$